

TUTEDUDE DEVOPS — ASSIGNMENT 4

Git & GitHub

Student: Umair Khan

Course: Tutedude DevOps

GitHub: <https://github.com/Umair-bey/tutedude-assignments>

Date: October 2026

TABLE OF CONTENTS

1. Environment Verification
2. Task 1 — Repository Setup & First Branch
3. Task 2 — Update JSON & Resolve Conflicts
4. Task 3 — Parallel Feature Development
5. Task 4 — Sequential Commits, Reset & Rebase
6. GitHub Repository
7. Conclusion

1. ENVIRONMENT VERIFICATION

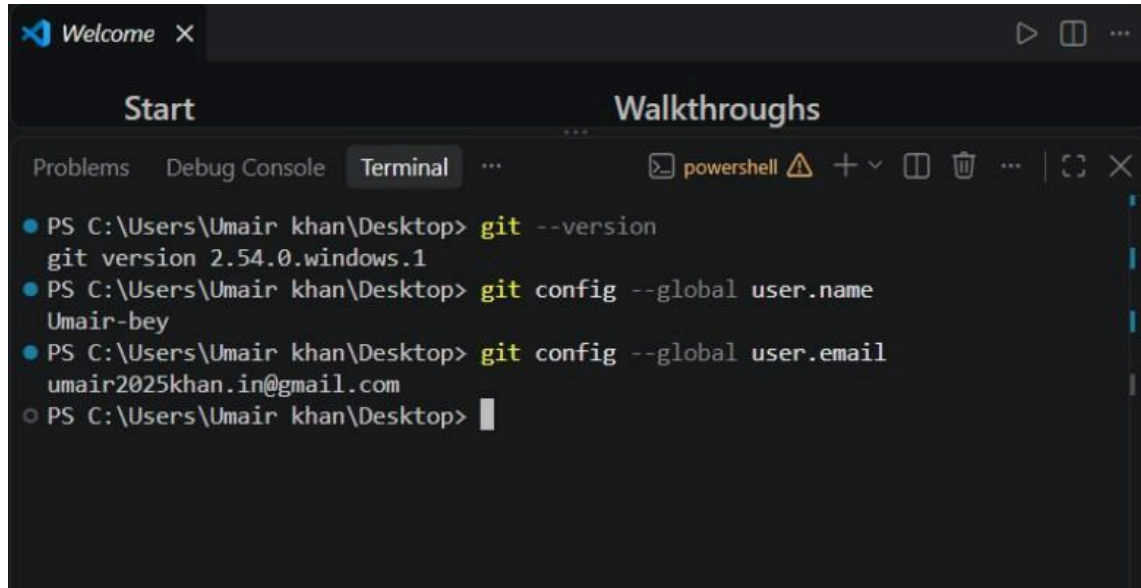
1.1 Git Version & Config

Commands:

```
git --version  
git config --global user.name  
git config --global user.email
```

Result:

```
Git version: 2.54.0.windows.1  
User: Umair-bey  
Email: umair2025khan.in@gmail.com
```



The image shows a Visual Studio Code interface with a terminal window open. The terminal is running PowerShell commands to install and configure Git. The output shows the Git version and the configuration of the user name and email.

```
Welcome X
Start Walkthroughs
Problems Debug Console Terminal ...
powershell
PS C:\Users\Umair khan\Desktop> git --version
git version 2.54.0.windows.1
PS C:\Users\Umair khan\Desktop> git config --global user.name
Umair-bey
PS C:\Users\Umair khan\Desktop> git config --global user.email
umair2025khan.in@gmail.com
PS C:\Users\Umair khan\Desktop>
```

1.2 Python, pip & PyMongo

Commands:

```
python --version
```

```
pip --version
```

```
python -c "import pymongo; print(pymongo._version_)"
```

Result:

```
Python: 3.14.2
```

```
pip: 25.3
```

```
PyMongo: 4.18.1
```

```
PS C:\Users\Umais khan\Desktop> python --version
Python 3.14.2
PS C:\Users\Umais khan\Desktop> pip --version
pip 25.3 from C:\Python314\Lib\site-packages\pip (python 3.14)
PS C:\Users\Umais khan\Desktop>

(.venv) PS C:\Users\Umais khan\Desktop> python -c "import pymongo; print(pymongo.__version__)"
4.18.1
(.venv) PS C:\Users\Umais khan\Desktop>
```

1.3 SSH Connection to GitHub

Command:

```
ssh -T git@github.com
```

Result:

Hi Umair-bey! You've successfully authenticated...

```
(.venv) PS C:\Users\Umais khan\Desktop\Flask_and_MongoDB_UmaisKhan> ssh -T git@github.com
The authenticity of host 'github.com (20.207.73.82)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wWV6TujhbpZisf/zLDA0zPMSvHdkr9UvCoqU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
Hi Umair-bey! You've successfully authenticated, but GitHub does not provide shell access.
(.venv) PS C:\Users\Umais khan\Desktop\Flask_and_MongoDB_UmaisKhan>
```

2. TASK 1 — REPOSITORY SETUP & FIRST BRANCH

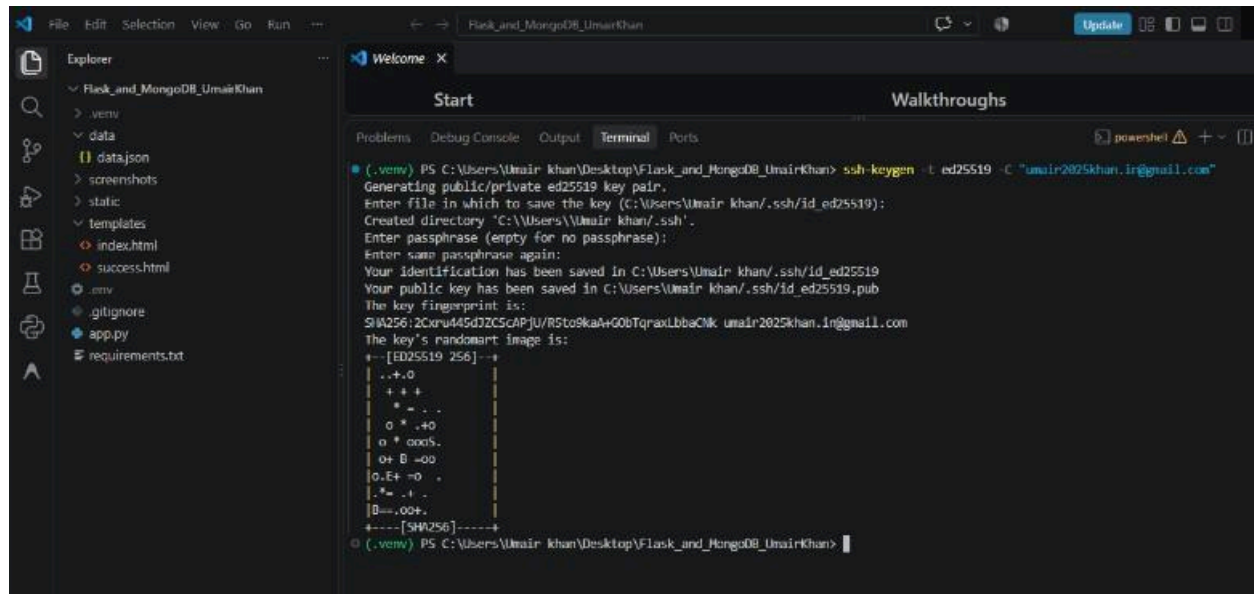
Step 1: SSH Key Generation

Command:

```
ssh-keygen -t ed25519 -C "umair2025khan.in@gmail.com"
```

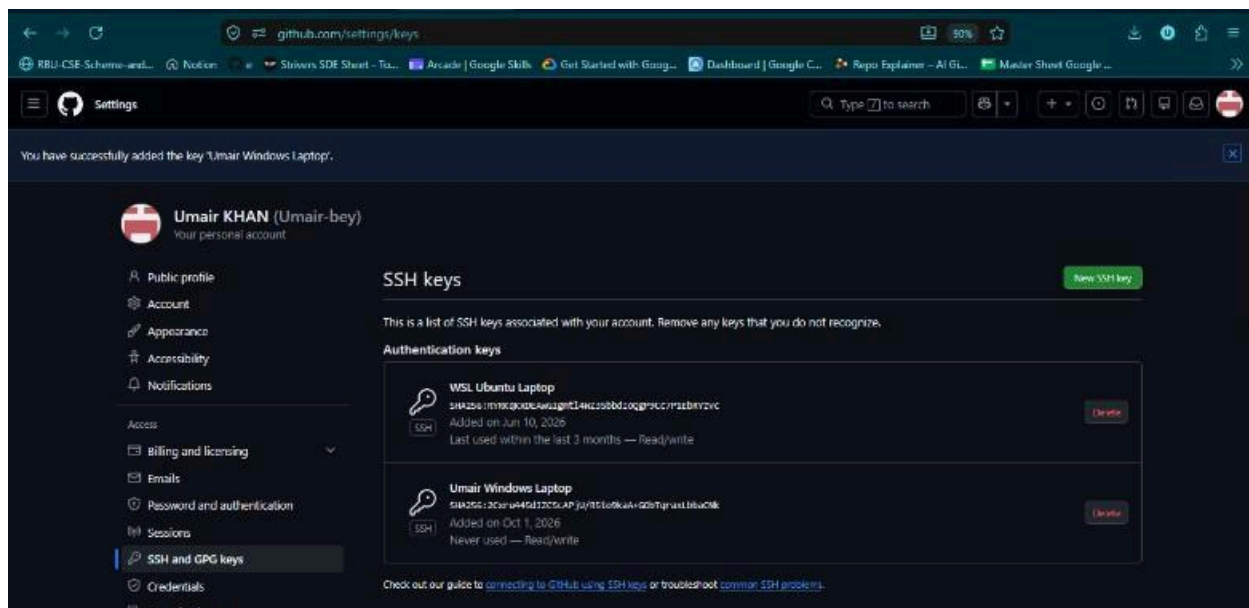
Result:

SSH key pair generated at ~/.ssh/id_ed25519 and id_ed25519.pub



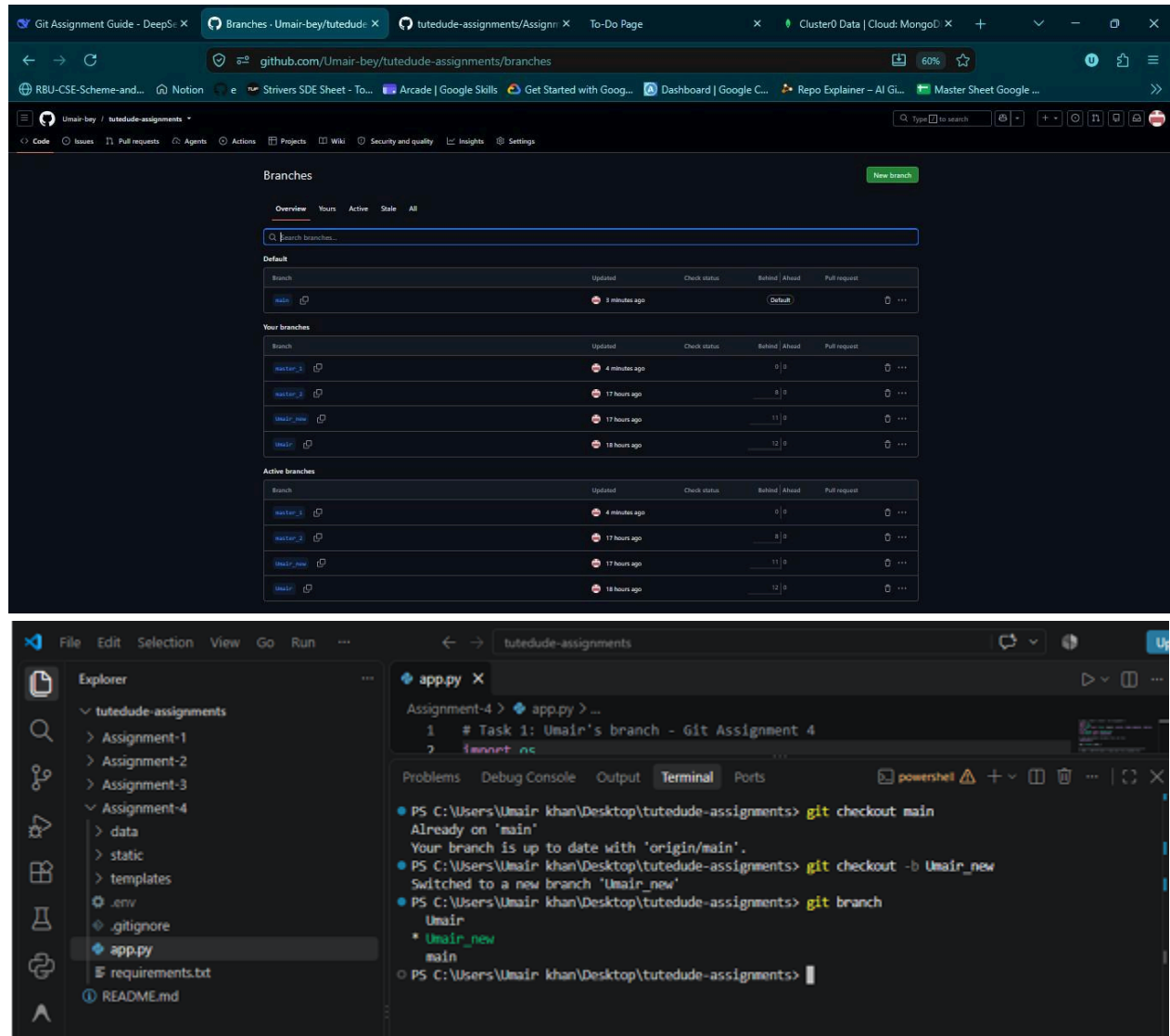
Step 2: SSH Key Added to GitHub

Key labeled "Umair Windows Laptop" added at github.com/settings/keys



Step 3: Branches Visible

Branches created: Umair, Umair_new, master_1, master_2, main



Step 4: Committed Change on Umair Branch

Commit: 600a2e1 — "Task 1: Add author comment to app.py in Umair branch"

Added author comment at the top of app.py

Step 5: Merged Umair into main

Commands:

```
git checkout main
git merge Umair
git push origin main
```

Result:

Fast-forward merge completed successfully

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Umais-bey/tutodude-assignments

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Notifications

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<> Code

Issues

Pull requests

Actions

Projects

Security and quality

Insights

Commits

History for tutodude-assignments / Assignment-4 on main

All users

All time

Commits on Oct 2, 2026

Task 4: Add Item UUID and Item Hash fields to To-Do form

Umais-bey committed 41 minutes ago

bebb1ab

<>

Task 4: Re-commit - state with only Item ID field in To-Do form

Umais-bey committed 1 hour ago

7f011ce

<>

Commits on Oct 1, 2026

Task 4: Add Item ID field to To-Do form

Umais-bey committed 15 hours ago

74212d1

<>

Task 3: Fix README filename typo (READNE -> README)

Umais-bey committed 15 hours ago

a8589a1

<>

Task 3: Add Assignment-4 README

Umais-bey committed 16 hours ago

64d839f

<>

Task 3: Properly resolve app.py conflict - keep both /todo and /submittodotitem routes

Umais-bey committed 17 hours ago

9c36605

<>

Task 3: Merge master_1 and master_2 - resolve conflicts, keep both routes

Umais-bey committed 17 hours ago

51b7836

<>

Task 3: Add /submittodotitem backend route with MongoDB storage

Umais-bey committed 17 hours ago

eedd11a

<>

Task 3: Add To-Do page with Item Name and Item Description fields

Umais-bey committed 17 hours ago

484c0e0

<>

Task 2: Update data.json in Umais_new branch

Umais-bey committed 18 hours ago

9f12c18

<>

Task 1: Add author comment to app.py in Umais branch

Umais-bey committed 18 hours ago

600a2e1

<>

Add Assignment 4: Flask project files with MongoDB Atlas setup

Umais-bey committed 18 hours ago

0cc2577

<>

End of commit history for this file

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3. TASK 2 — UPDATE JSON & RESOLVE CONFLICTS

=====

Step 1: Created Branch Umais_new

Command:

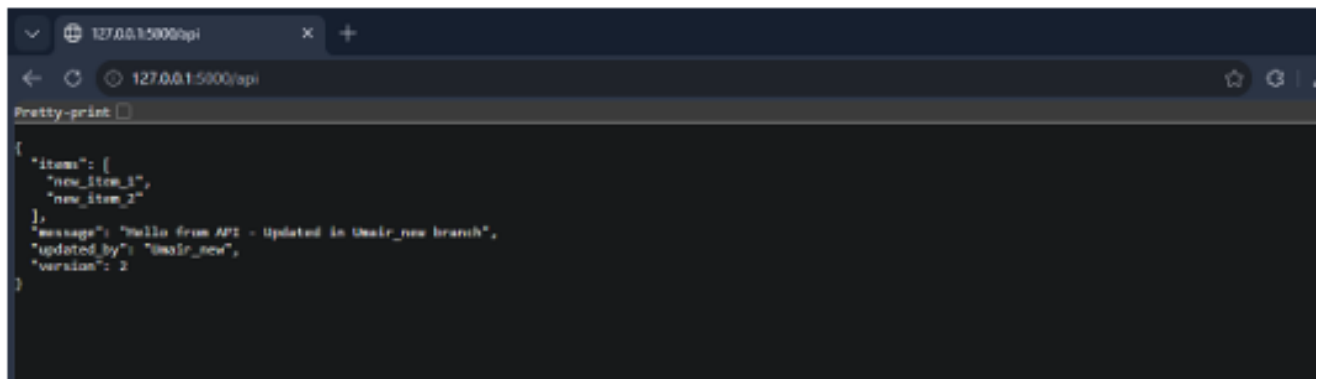
```
git checkout -b Umair_new
```

Step 2: Updated Assignment-4/data/data.json on Umair_new

New content:

```
{
  "message": "Hello from API - Updated in Umair_new branch",
  "version": 2,
  "items": ["new_item_1", "new_item_2"],
  "updated_by": "Umair_new"
}
```

Commit: 9f12c28 — "Task 2: Update data.json in Umair_new branch"



Step 3: Modified data.json on main Differently

New content:

```
{
  "message": "Hello from API - Updated in main branch",
  "version": 3,
  "items": ["main_item_1"],
  "updated_by": "main"
}
```

Commit: 2a8f742 — "Task 2: Update data.json in main branch"

Step 4: Merged Umair_new → main (CONFLICT)

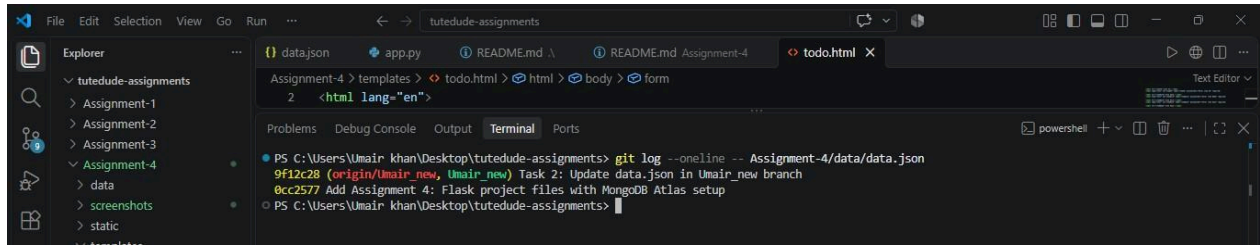
Command:

```
git merge Umair_new
```

Output:

CONFLICT (content): Merge conflict in Assignment-4/data/data.json

Automatic merge failed; fix conflicts and then commit the result.



Step 5: Conflict Markers & Resolution

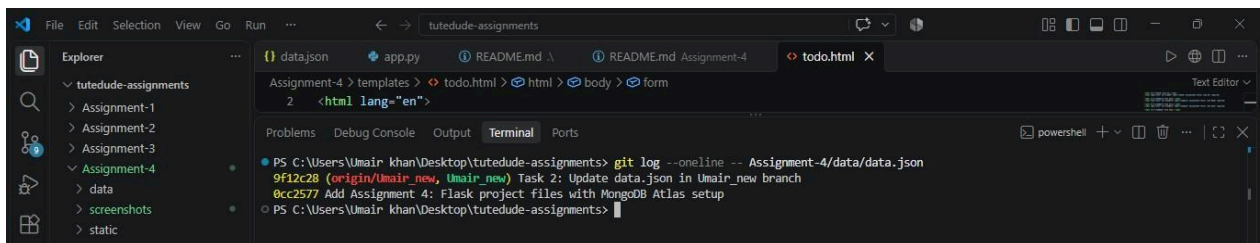
Conflict markers appeared in data.json:

```
<<<<<<< HEAD
{...main version...}
=====
{...Umair_new version...}
>>>>>>> Umair_new
```

Resolution: Accepted the Umair_new version (removed all markers)

Step 6: Staged, Committed, Pushed

Commit: 053866f — "Task 2: Resolve merge conflict - accept changes from Umair_new"



4. TASK 3 — PARALLEL FEATURE DEVELOPMENT

Step 1: Created Two Branches from main

Commands:

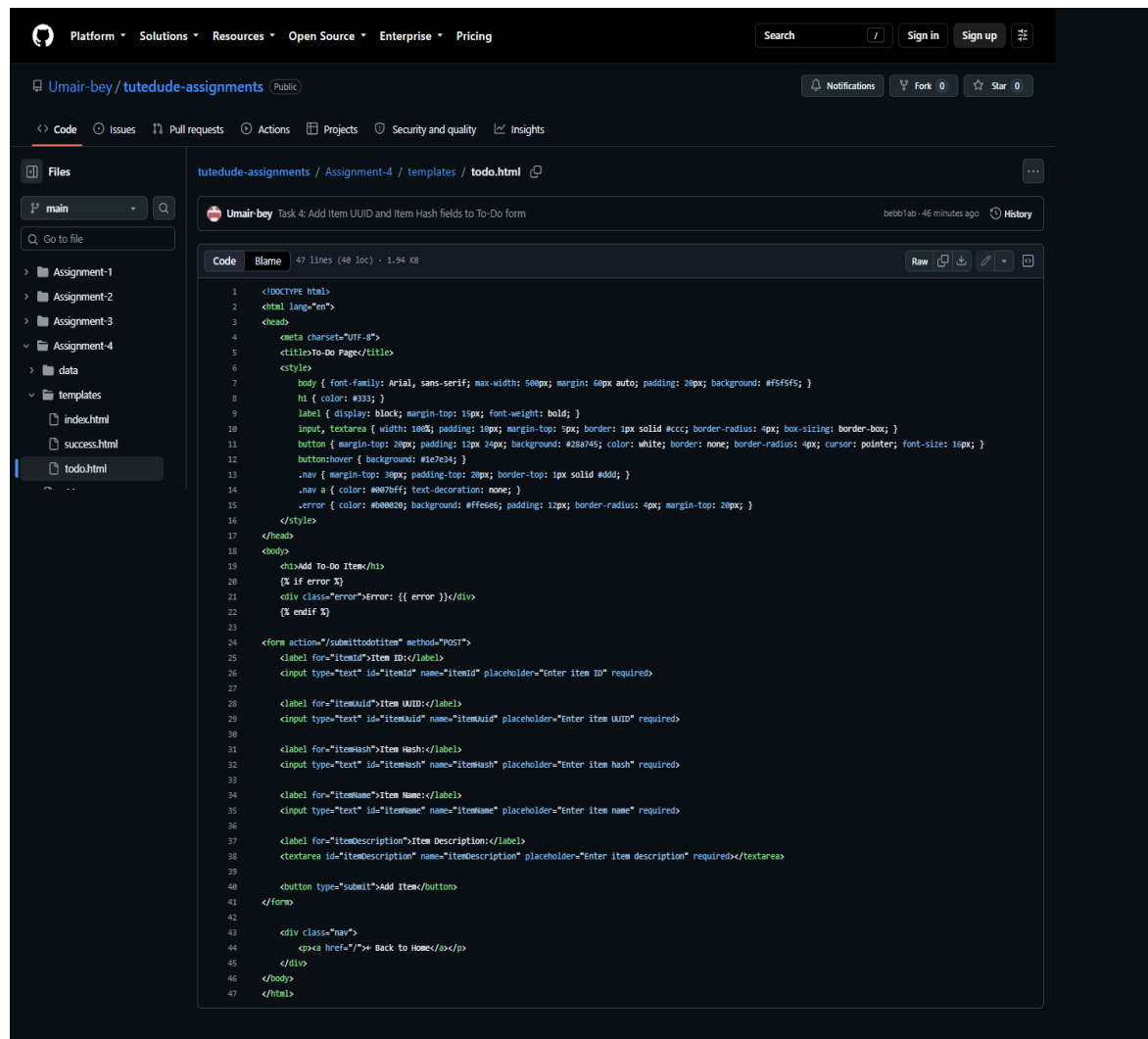
```
git checkout main
```

```
git checkout -b master_1
git checkout main
git checkout -b master_2
```

Step 2: Frontend To-Do Page in master_1

File: Assignment-4/templates/todo.html

Commit: 4d4c0e0 — "Task 3: Add To-Do page with Item Name and Item Description fields"

The screenshot shows a GitHub repository named 'tutodude-assignments' by 'Ulmair-bey'. The file 'Assignment-4 / templates / todo.html' is selected. The commit history shows 'Task 4: Add Item UUID and Item Hash fields to To-Do form' by 'beb01ab' 46 minutes ago. The code is a single file, 47 lines (48 loc), 1.94 KB. The code is a mix of HTML, CSS, and JavaScript. It includes a form with fields for 'Item ID', 'Item UUID', 'Item Hash', 'Item Name', and 'Item Description'. The form has a 'Submit' button. The code also includes a 'Back to Home' link. The CSS is inline, defining styles for the form elements and the body. The JavaScript is also inline, handling the form submission and error handling. The code is as follows:

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <title>To-Do Page</title>
6   <style>
7     body { font-family: Arial, sans-serif; max-width: 500px; margin: 60px auto; padding: 20px; background: #f5f5f5; }
8     h1 { color: #333; }
9     label { display: block; margin-top: 15px; font-weight: bold; }
10    input, textarea { width: 100%; padding: 10px; margin-top: 5px; border: 1px solid #ccc; border-radius: 4px; box-sizing: border-box; }
11    button { margin-top: 20px; padding: 12px 24px; background: #21a745; color: white; border: none; border-radius: 4px; cursor: pointer; font-size: 16px; }
12    button:hover { background: #1e7e34; }
13    .nav { margin-top: 30px; padding-top: 20px; border-top: 1px solid #ddd; }
14    .nav a { color: #007bff; text-decoration: none; }
15    .error { color: #000000; background: #ffe0e0; padding: 12px; border-radius: 4px; margin-top: 20px; }
16  </style>
17 </head>
18 <body>
19   <h1>Add To-Do Item</h1>
20   <% if error %>
21   <div class="error">error: {{ error }}</div>
22   <% endif %>
23
24   <form action="/submittodotitem" method="POST">
25     <label for="itemid">Item ID:</label>
26     <input type="text" id="itemid" name="itemid" placeholder="Enter item ID" required>
27
28     <label for="itemuid">Item UUID:</label>
29     <input type="text" id="itemuid" name="itemuid" placeholder="Enter item UUID" required>
30
31     <label for="itemhash">Item Hash:</label>
32     <input type="text" id="itemhash" name="itemhash" placeholder="Enter item hash" required>
33
34     <label for="itemname">Item Name:</label>
35     <input type="text" id="itemname" name="itemname" placeholder="Enter item name" required>
36
37     <label for="itemdescription">Item Description:</label>
38     <textarea id="itemdescription" name="itemdescription" placeholder="Enter item description" required></textarea>
39
40     <button type="submit">Add Item</button>
41   </form>
42
43   <div class="nav">
44     <a href="/">Back to Home</a></div>
45 </body>
46 </html>
```

Step 3: Backend API in master_2

Added /submittodotitem POST route to Assignment-4/app.py

Routes visible: /api, /submit, /success, /todo, /submittodotitem

Commit: eeddd1a — "Task 3: Add /submittodotitem backend route with MongoDB storage"

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Files

main

Go to file

Assignment-1

Assignment-2

Assignment-3

Assignment-4

data

templates

.gitignore

README.md

app.py

tutode-assignments / Assignment-4 / app.py

Umair-bey Task 3: Properly resolve app.py conflict - keep both /todo and /submi... 9c3605 · 17 hours ago

CodeBlame88 lines (70 loc) · 2.61 KB

RawDownloadEditView

```
1 # Task 1: Umair's branch - Git Assignment 4
2 import os
3 import json
4 from flask import Flask, request, jsonify, render_template, redirect, url_for
5 from pymongo import MongoClient
6 from dotenv import load_dotenv
7
8 # Load environment variables from .env file
9 load_dotenv()
10
11 app = Flask(__name__)
12
13 # =====
14 # TASK 1: JSON API Route - reads data from a backend file
15 # =====
16 DATA_FILE = os.path.join(os.path.dirname(__file__), 'data', 'data.json')
17
18 def read_data_file():
19     """Read the JSON list from the backend file."""
20     with open(DATA_FILE, 'r') as f:
21         return json.load(f)
22
23 @app.route('/api', methods=['GET'])
24 def api():
25     """Return the JSON list stored in data.json"""
26     data = read_data_file()
27     return jsonify(data)
28
29 # =====
30 # TASK 2: Frontend Form + MongoDB Atlas
31 # =====
32 MONGO_URI = os.getenv("MONGO_URI")
33 client = MongoClient(MONGO_URI)
34 db = client.get_database('tutode_db')
35 collection = db['submissions']
36
37
38 @app.route('/')
39 def index():
40     """Show the form."""
41     return render_template('index.html')
42
43
44 @app.route('/submit', methods=['POST'])
45 def submit():
46     """Handle form submission."""
47     try:
48         data = request.form.to_dict()
49         if not data.get('name') or not data.get('email'):
50             raise ValueError("Name and Email are required.")
51         collection.insert_one(data)
52         # Success: redirect to a different page
53         return redirect(url_for('success'))
54     except Exception as e:
55         # Error: render the same page with the error message
56         return render_template('index.html', error=str(e))
57
58
59 @app.route('/success')
60 def success():
61     """Success page shown after form submission."""
62     return render_template('success.html')
63
64
65 @app.route('/todo')
66 def todo():
67     return render_template('todo.html')
68 @app.route('/submittodoitem', methods=['POST'])
69 def submit_todo_item():
70     """Handle To-Do form submission and store in MongoDB."""
71     try:
72         item_name = request.form.get('itemName')
73         item_description = request.form.get('itemDescription')
74
75         if not item_name or not item_description:
76             raise ValueError("Item Name and Item Description are required.")
77
78         collection.insert_one({
79             'itemName': item_name,
80             'itemDescription': item_description
81         })
82
83         return redirect(url_for('todo'))
84     except Exception as e:
85         return render_template('todo.html', error=str(e))
86
87 if __name__ == '__main__':
88     app.run(debug=True)
```

Symbols

Find definitions and references for function symbols in this file by clicking a symbol or code.

Filter symbols

Step 4: Merged Both Branches into main

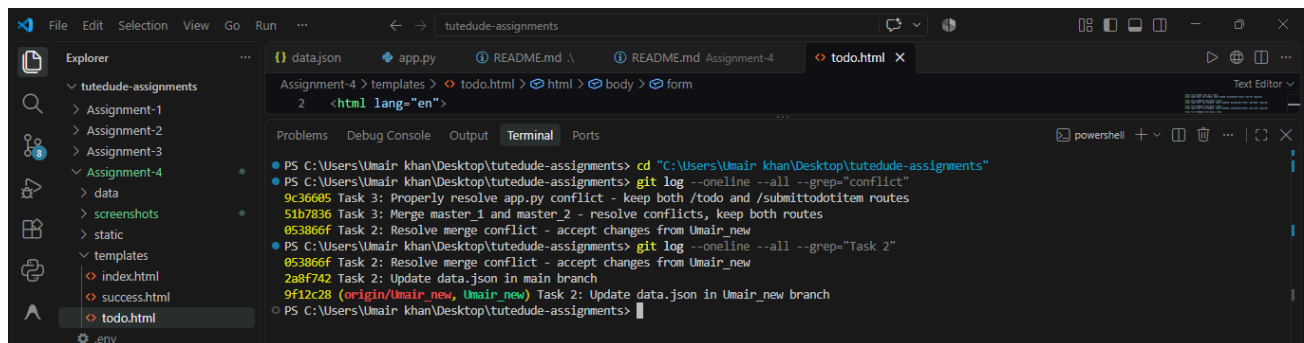
Commands:

```
git checkout main
git merge master_1
git merge master_2
```

Conflict in app.py (both branches added routes)

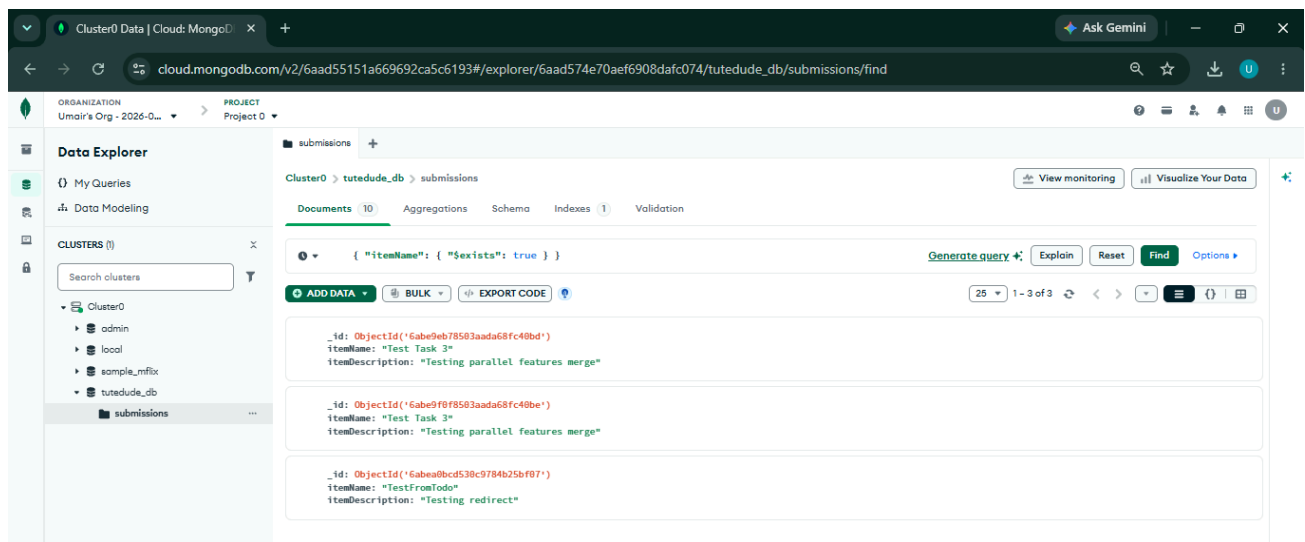
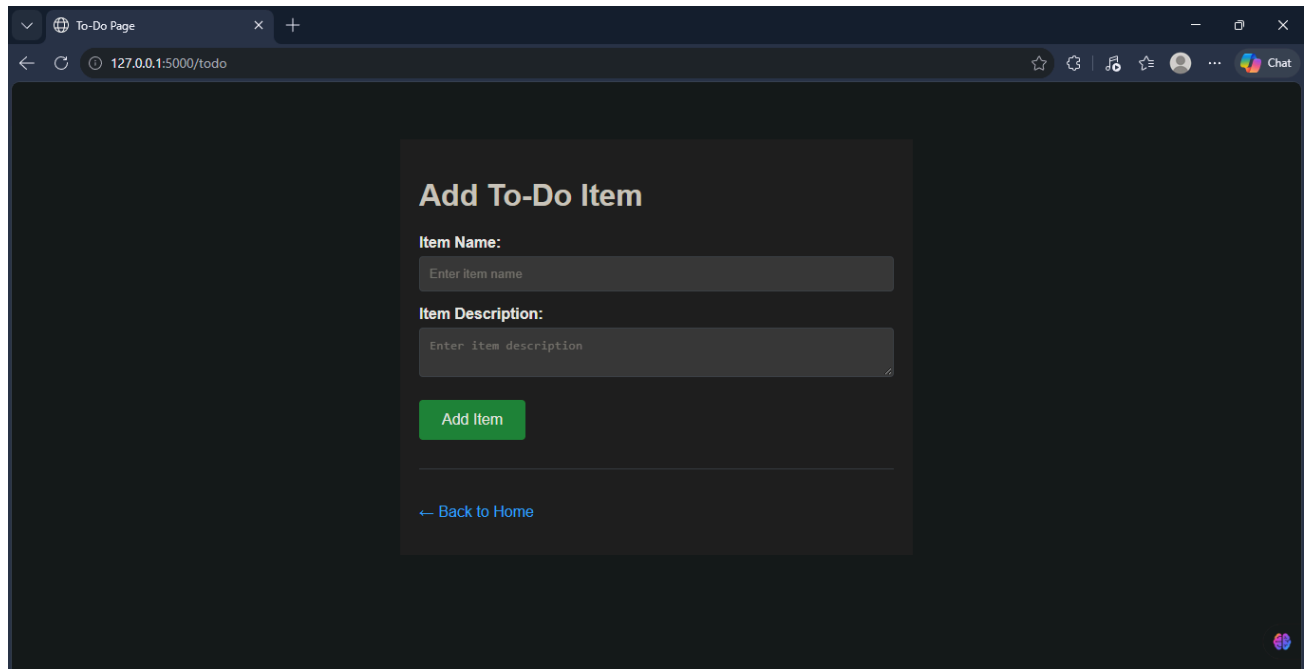
Resolution: Accepted both changes — kept /todo AND /submittodotitem

Commit: 51b7836 — "Task 3: Merge master_1 and master_2 - resolve conflicts, keep both routes"



Step 5: Tested To-Do Form End-to-End

- Flask server running on `http://localhost:5000`
- Submitted To-Do form
- Data stored in MongoDB Atlas collection `tutodude_db.submissions`



Step 6: Verified Assignment-4 Folder on GitHub

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Notifications Fork 0 Star 0

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Files

main

Go to file

Assignment-1

Assignment-2

Assignment-3

Assignment-4

data

templates

.gitignore

README.md

app.py

.

tutedude-assignments / Assignment-4 /

Umais-bey Task 4: Add Item UUID and Item Hash fields to To-Do form bebb1ab · 43 minutes ago History

Name	Last commit message	Last commit date
..		
data	Task 2: Update data.json in Umais_new branch	18 hours ago
templates	Task 4: Add Item UUID and Item Hash fields to To-Do form	43 minutes ago
.gitignore	Add Assignment 4: Flask project files with MongoDB Atlas setup	18 hours ago
README.md	Task 3: Fix README filename typo (READNE -> README)	15 hours ago
app.py	Task 3: Properly resolve app.py conflict - keep both /todo and /submi...	17 hours ago
requirements.txt	Add Assignment 4: Flask project files with MongoDB Atlas setup	18 hours ago

README.md

Assignment 4: Git & GitHub

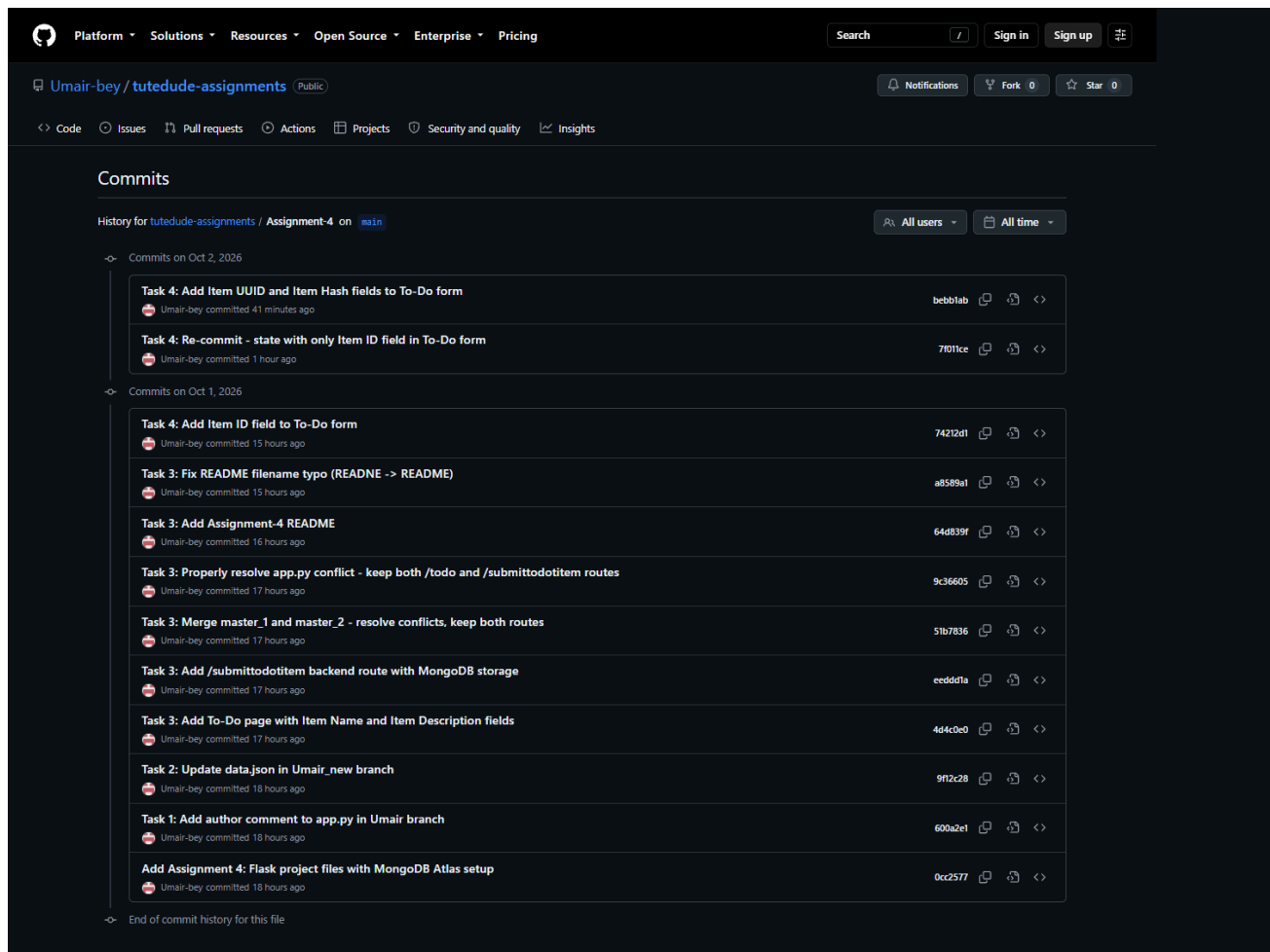
Author: Umais Khan Course: Tutedude DevOps Repo: <https://github.com/Umais-bey/tutedude-assignments>

Tasks Completed

- Task 1: Repository setup, SSH key, branch Umais
- Task 2: JSON update and merge conflict resolution
- Task 3: Parallel branches master_1 and master_2 with MongoDB Atlas
- Task 4: Sequential commits, soft reset, rebase

Routes

- / — Home form
- /api — JSON data
- /todo — To-Do form
- /submittodotitem — POST to MongoDB



=====

5. TASK 4 — SEQUENTIAL COMMITS, RESET & REBASE

=====

Step 1: Synced master_1 with main

Command:

```
git checkout master_1
```

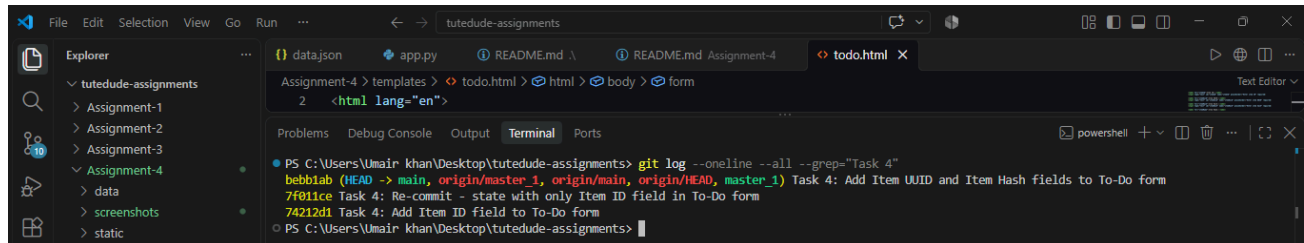
git merge main

Step 2: Added 3 Fields Sequentially (3 Separate Commits)

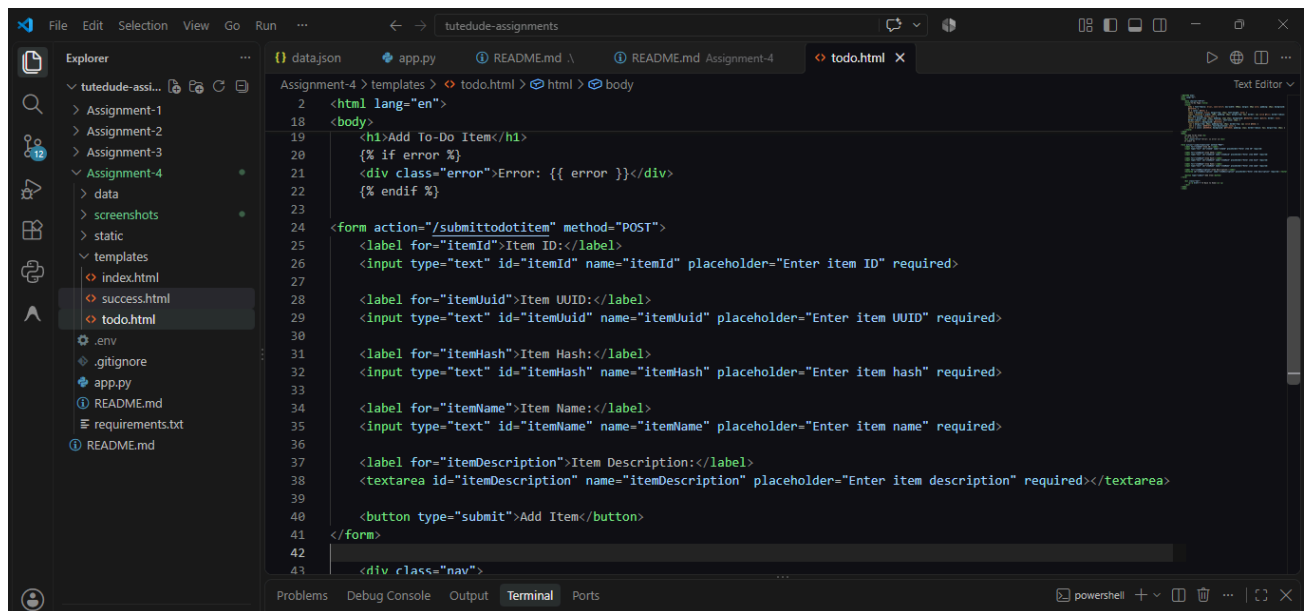
Commit 1 — 74212d1 — "Task 4: Add Item ID field to To-Do form"

Commit 2 — be02f52 — "Task 4: Add Item UUID field to To-Do form"

Commit 3 — cf2bc93 — "Task 4: Add Item Hash field to To-Do form"



```
PS C:\Users\Umair Khan\Desktop\tutodude-assignments> git log --oneline --all --grep="Task 4"
bebb1ab (HEAD -> main, origin/master 1, origin/main, origin/HEAD, master 1) Task 4: Add Item UUID and Item Hash fields to To-Do form
7f011ce Task 4: Re-commit - state with only Item ID field in To-Do form
74212d1 Task 4: Add Item ID field to To-Do form
```



```
2 <html lang="en">
18 <body>
19 <h1>Add To-Do Item</h1>
20 {% if error %}
21 <div class="error">Error: {{ error }}</div>
22 {% endif %}
23
24 <form action="/submittodotitem" method="POST">
25 <label for="itemId">Item ID:</label>
26 <input type="text" id="itemId" name="itemId" placeholder="Enter item ID" required>
27
28 <label for="itemUuid">Item UUID:</label>
29 <input type="text" id="itemUuid" name="itemUuid" placeholder="Enter item UUID" required>
30
31 <label for="itemHash">Item Hash:</label>
32 <input type="text" id="itemHash" name="itemHash" placeholder="Enter item hash" required>
33
34 <label for="itemName">Item Name:</label>
35 <input type="text" id="itemName" name="itemName" placeholder="Enter item name" required>
36
37 <label for="itemDescription">Item Description:</label>
38 <textarea id="itemDescription" name="itemDescription" placeholder="Enter item description" required></textarea>
39
40 <button type="submit">Add Item</button>
41 </form>
42
43 <div class="nav">
```

Final todo.html contains all 5 fields:

Item ID, Item UUID, Item Hash, Item Name, Item Description

Step 3: Merged master_1 into main

Command:

git checkout main

git merge master_1

Step 4: Soft Reset to Item ID Commit

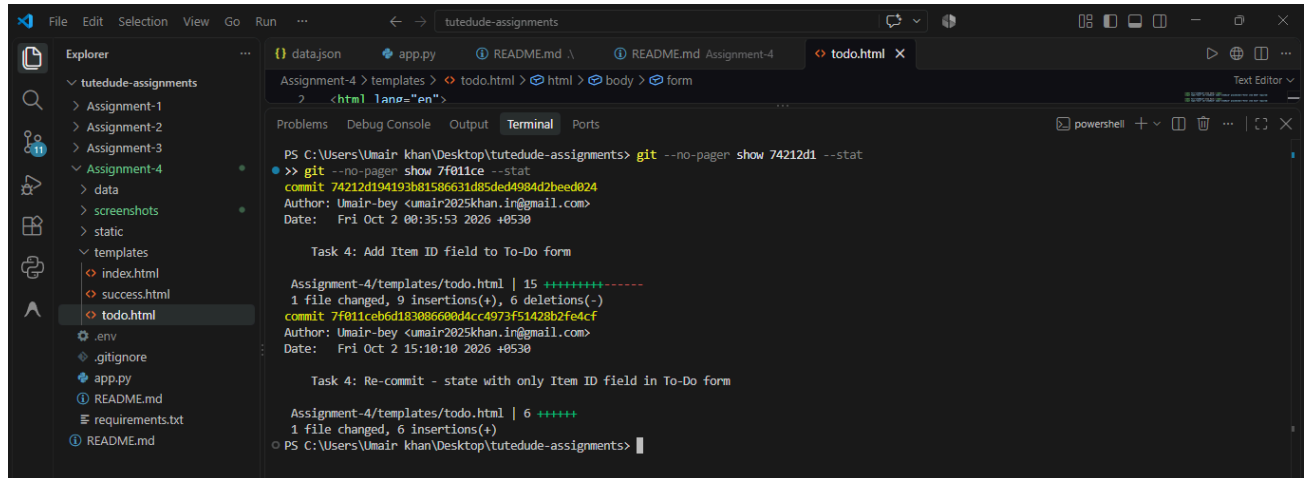
Command:

git reset --soft 74212d1

Result:

UUID and Hash changes became staged but not committed

Re-committed: 7f011ce — "Task 4: Re-commit - state with only Item ID field in To-Do form"



The screenshot shows the Visual Studio Code interface with a terminal window open. The terminal displays the output of a git commit command. The commit message is "Task 4: Re-commit - state with only Item ID field in To-Do form". The commit hash is 7f011ceb6d18308660d4cc4073f51428b2fe4cf. The terminal also shows the output of a git log command, displaying the commit history. The commit 7f011ce is highlighted in the log output.

```
PS C:\Users\Umair_khan\Desktop\tutode-assignments> git --no-pager show 74212d1 --stat
>> git --no-pager show 7f011ce --stat
commit 74212d194193b81586631d85ded4984d2beed024
Author: Umair-bey <umair2025khan.ir@gmail.com>
Date: Fri Oct 2 00:35:53 2026 +0530

    Task 4: Add Item ID field to To-Do form

Assignment-4/templates/todo.html | 15 ++++++-----
1 file changed, 9 insertions(+), 6 deletions(-)
commit 7f011ceb6d18308660d4cc4073f51428b2fe4cf
Author: Umair-bey <umair2025khan.ir@gmail.com>
Date: Fri Oct 2 15:10:10 2026 +0530

    Task 4: Re-commit - state with only Item ID field in To-Do form

Assignment-4/templates/todo.html | 6 +++++
1 file changed, 6 insertions(+)
```

Shows difference between original Item ID commit (74212d1) and re-commit (7f011ce)

Step 5: Rebase master_1 onto Updated main

Command:

```
git checkout master_1
git rebase main
```

Conflict in todo.html — resolved by keeping all 5 fields
Result: Successfully rebased and updated refs/heads/master_1

Step 6: Force-Pushed master_1

Command:

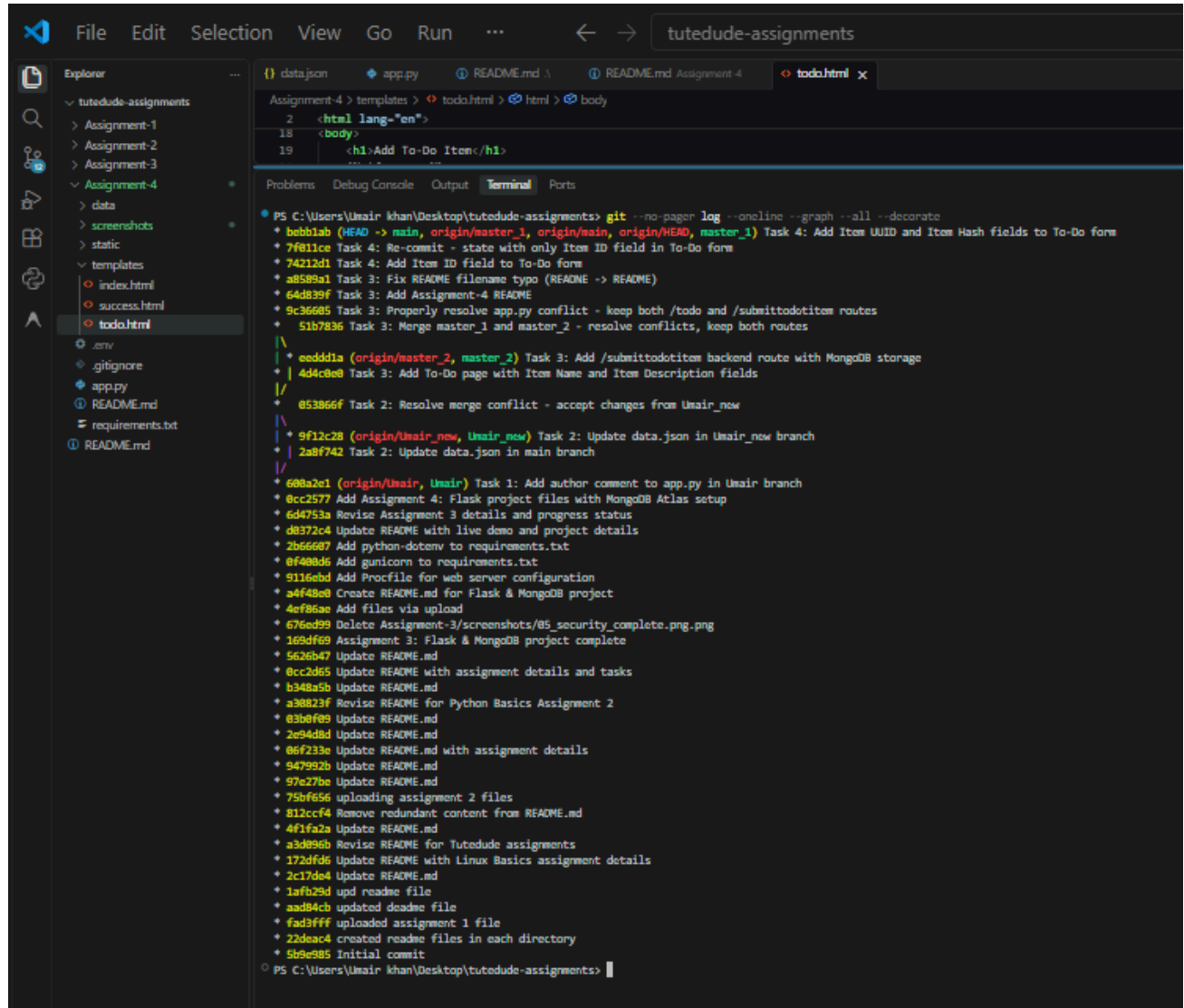
```
git push origin master_1 --force
```

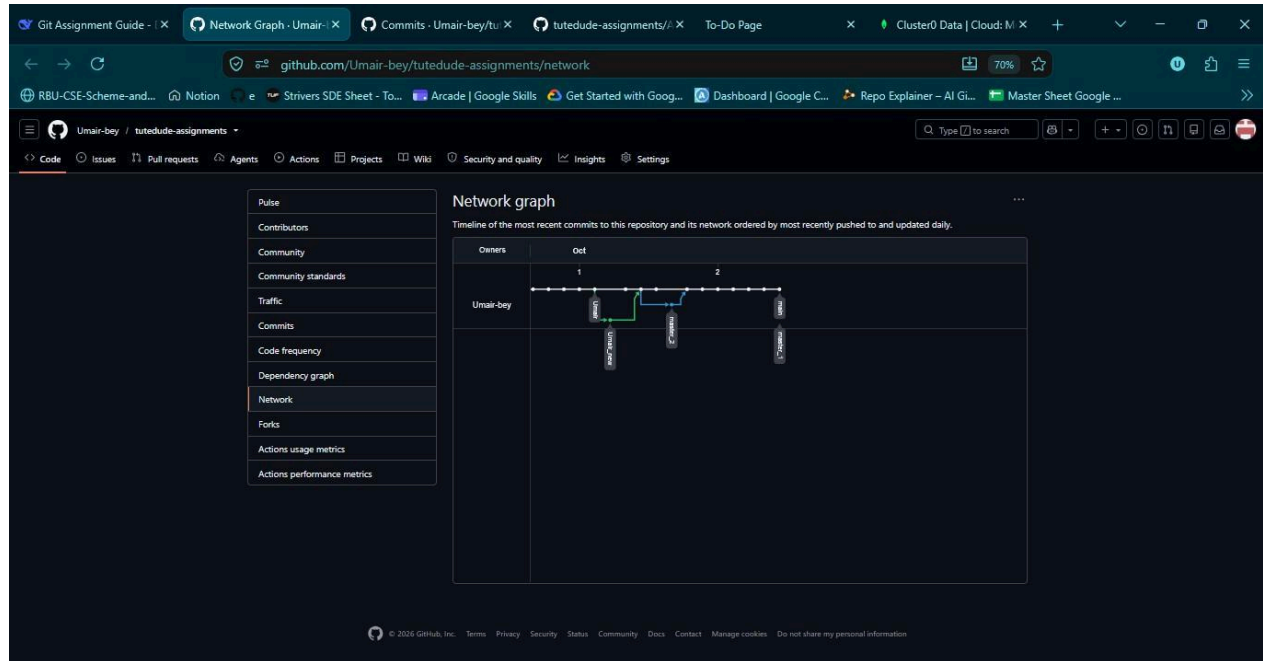
Step 7: Merged Final State into main

Commands:

```
git checkout main  
git merge master_1  
git push origin main
```

Step 8: Final Commit Graph





Confirms: sequential commits, reset, and rebase are all visible in history

=====

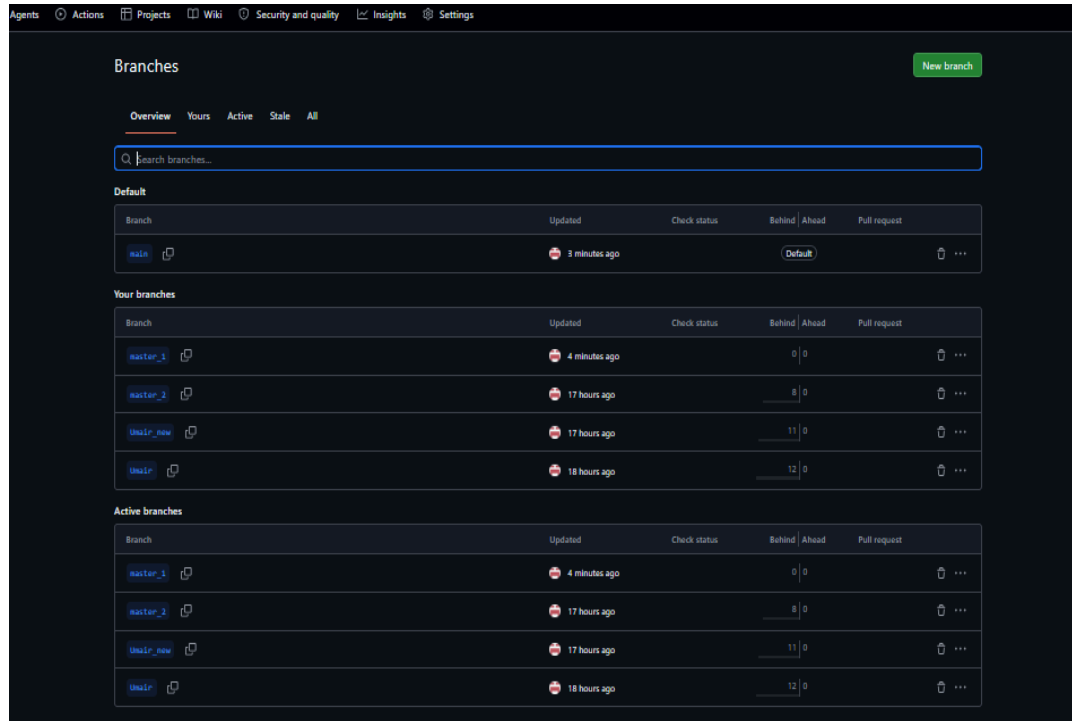
6. GITHUB REPOSITORY

=====

Repository: <https://github.com/Umais-bey/tutedude-assignments>

Assignment 4 folder:

<https://github.com/Umais-bey/tutedude-assignments/tree/main/Assignment-4>



=====

6.5 LIVE DEPLOYMENT

=====

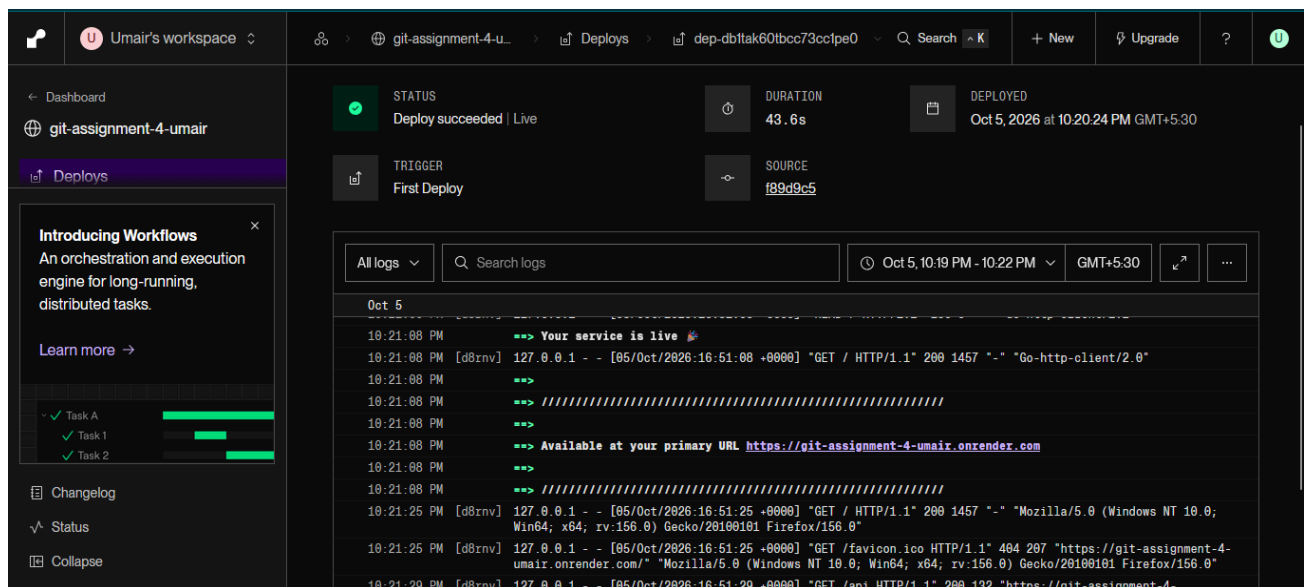
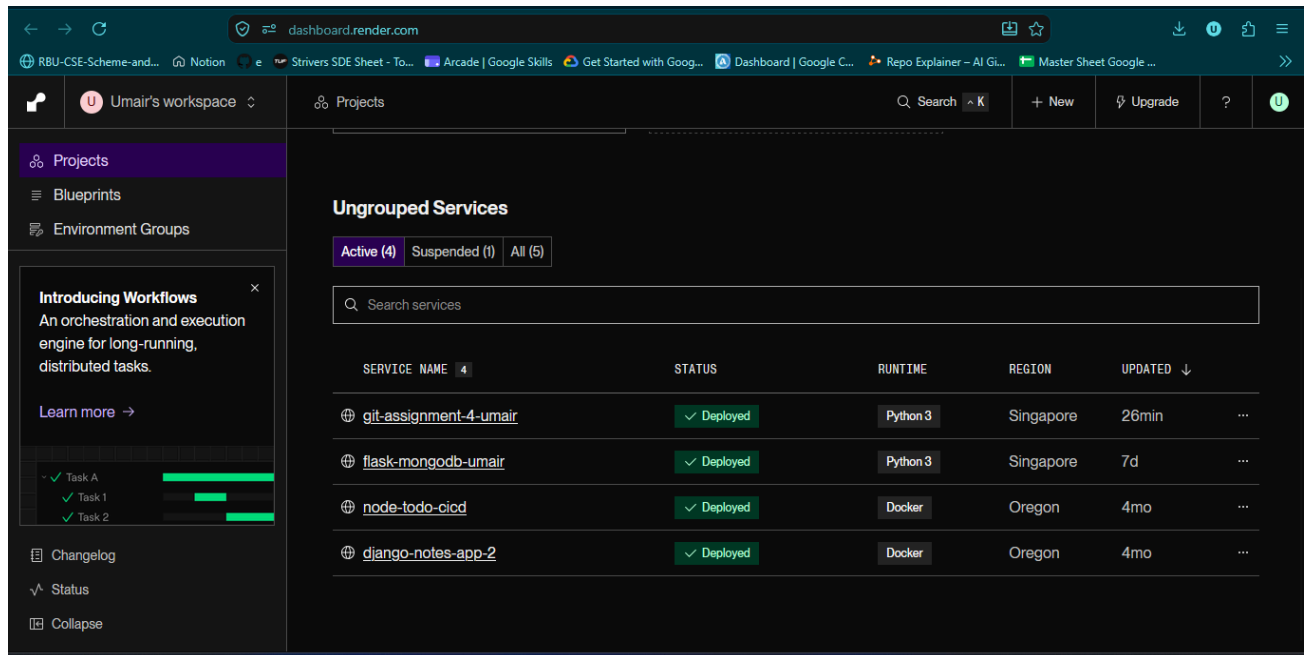
The Flask application was deployed to production using Render.com.

DEPLOYED URL:

<https://git-assignment-4-umair.onrender.com>

CONFIGURATION:

- Platform: Render.com (Free tier)
- Root Directory: Assignment-4
- Build Command: `pip install -r requirements.txt`
- Start Command: `gunicorn app:app`
- Environment Vars: `MONGO_URI`, `FLASK_DEBUG=False`



LIVE TESTING:

- ✓ Home page — loads successfully
- ✓ /todo — To-Do form with all 5 fields works
- ✓ /api — returns JSON data
- ✓ To-Do submissions save to MongoDB Atlas

https://git-assignment-4-umair.onrender.com/todo

Add To-Do Item

Item ID:
LIVE001

Item UUID:
uuid-live-001

Item Hash:
hash-live-001

Item Name:
Live Deploy Test

Item Description:
Testing live deployment

Add Item

cloud.mongodb.com/v2/6aad55151a669692ca5c6193#/explorer/6aad574e70aef6908dafc074/tutedude_db/submissions

ORGANIZATION: Umair's Org - 2026-0... PROJECT: Project 0

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Cluster0

- admin
- local
- sample_mflix
- tutedude_db
 - submissions

submissions

Cluster0 > tutedude_db > submissions

Documents 11 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain Reset Find Options

ADD DATA BULK EXPORT CODE

25 1 - 11 of 11

itemName: "Test Task 3"	itemDescription: "Testing parallel features merge"	
_id: ObjectId('6abe9f8f8583aada68fc48be')	itemName: "Test Task 3"	itemDescription: "Testing parallel features merge"
_id: ObjectId('6abea0bcd530c9784b25bf07')	itemName: "TestFromTodo"	itemDescription: "Testing redirect"
_id: ObjectId('6ac3dd2b05c2a198564b9876')	itemName: "Live Deploy Test"	itemDescription: "Submitted from Render deployment"

Alert System Status: All Good

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Note: Render's free tier sleeps after 15 min of inactivity.

First request after sleep takes 30–60 seconds.

7. CONCLUSION

All 4 tasks completed successfully:

Task 1: SSH setup, branch Umair created and merged to main

Task 2: Branch Umair_new, JSON conflict resolved by accepting _new changes

Task 3: Parallel branches master_1 (frontend) + master_2 (backend API),

merged with MongoDB Atlas integration

Task 4: Sequential commits (ID → UUID → Hash), soft reset, rebase,
and final merge

PROJECT STRUCTURE

Assignment-4/

```
|— app.py
|— requirements.txt
|— .env          (gitignored — contains MongoDB URI)
|— .gitignore
|— data/
|   |— data.js
```



```
|— templates/
|   |— index.html
|   |— success.html
|   |— todo.html
|— static/
|— screenshots/
|   |— [all task screenshots]
|— README.md
```

- GitHub Repository: <https://github.com/Umair-bey/tutedude-assignments>